VIT-AP UNIVERSITY

## QUESTION PAPER

## Name of the Examination: WIN 2023-24 Semester– CAT1

**Course Code:** CSE3003

**Course Title:** Computer Networks

**Set number:** 2

**Date of Exam:** 05/02/2024 (AN)

**Duration:** 90mins

**Total Marks:** 50

(A2)

**Instructions:**

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

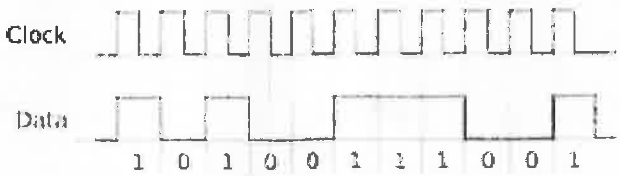
Q1. In the network when a data transmission is done, identify the layer(s) which acts as Dialog controller and encoding. Explain the functions of that layer in the OSI model along with the protocols used in it. Also, explain the layer in TCP/IP which does that function. (10M)

Q2. What are the factors that determine whether a communication system is a LAN or WAN? An institution wants to deploy a high-speed network with two internetworking within a 2km radius. Suggest the network type, and minimum number of network components to establish this high-speed network. Explain in detail. (10M)

Q3. Interpret which type of transmission transmits data through physical cables, while the other relies on wireless signals for transmission. Discuss the advantages of it. Identify the type of media used in Ethernet LAN cables within homes and offices that can transport up to 10 Gbit/s Ethernet. (10M)

Q4. Discuss the consequences for different types of network topology (Bus, Star, Ring and Mesh) if a connection fails. Among the four which is robust, does not incapacitate the entire system, provides privacy or security and every message travels along a dedicated line, only the intended recipient sees it. Explain. (10M)

Q5. A digital bit stream consisting of the following bits given in the figure is to be transmitted. Draw the following waveforms for the transmitted data (10M)




a) **Manchester**

b) **Differential Manchester**

c) **Polar NRZ-I**

d) **Polar NRZ-L**

e) **Pseudoternary**

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## QUESTION PAPER

### Name of the Examination: WIN 2023-24 Semester – CAT-1

**Course Code:** CSE3003

**Course Title:** Computer Networks

**Set number:** 4

**Date of Exam:** 06/02/2024 (AN) (B2)

**Duration:** 90 minutes

**Total Marks:** 50

**Instructions:**

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

**Q1.** Discuss the characteristics of the following network components briefly.

**(a)** Hub **(b)** Switches **(c)** Bridges **(d)** Routers and **(e)** Gateways **(10M)**

**Q2.** **(a)** In a networking layer device, what is the purpose of having physical layer and datalink layer ?

Justify with suitable reasons. **(5M)**

**(b)** Explain the scenario of whether the segment encapsulates frames or frames encapsulate segments. How is error control at transport layer different from error control at datalink layer ? **(5M)**

**Q3.** **(a)** What are known issues in the unguided transmission media? Why generally radio waves are used in cellular communication not microwave? **(5M)**

**(b)** Differentiate between distortion and attenuation. Identify and discuss the properties of the transmission media that is least affected by the above impairments. **(5M)**

**Q4.** **(a)** Can we convert a star topology comprising of five computers and a hub into a dual ring topology ? How is network setup cost and scalability affected ? Justify. **(5M)**

**(b)** Illustrate a topology that has only two neighbours for each device. Write the pros and cons of the given topology. **(5M)**

**Q5.** Encode the binary data 101000110 using the following encoding methods. **(2.5 x 4 M = 10M)**

a. Non-Return Zero (NRZ)-L
b. Non-Return Zero (NRZ)-I
c. Differential Manchester
d. Return Zero (RZ)

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## QUESTION PAPER

## Name of the Examination: WINTER Semester 2023-24 – CAT 1

**Course Code: CSE3003** | **Course Title: Computer Networks**

**Set number:** 6 | **Date of Exam:** 05/02/2024 (FN) (A1)

**Duration: 90 Min** | **Total Marks: 50**

**Instructions:**

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

**Q1.** Discuss the significance of layering in the OSI model for facilitating network communication. Explain process to process delivery transport layer. **(10M)**

**Q2.** a. How does the choice of network architecture impact scalability, reliability, and performance in networking environment? **(5M)**

b. Switch is usually a unicast device but sometimes it broadcast also. Justify with suitable reasons. **(5M)**

**Q3.** Distinguish between guided media and unguided media. In physical layer what are different types of energy that are transmitted and which type of transmission media used to carry them? **(10M)**

**Q4.** a. List out all the features, advantages and disadvantages of a point-to-point network topology? **(5M)**

b. Consider a small office with five computers connected to each other. The office manager wants to set up a network where each computer can communicate directly with any another computer. Which network architecture would be the most suitable for this scenario? Justify your answer with a neat sketch. **(5M)**

**Q5.** Apply NRZ-L, NRZ-I, RZ, Manchester, and Differential Manchester encoding techniques to the binary data "10101100011" and draw the corresponding waveforms. **(10M)**

**QP MAPPING**

| Q. No. | Module Number | CO Mapped | PO Mapped | PEO Mapped | PSO Mapped | Marks |
|---|---|---|---|---|---|---|
| Q1 | 1 | 2 | 1 | 1 | 1 | 10 |
| Q2 | 1 | 1 | 1 | 1 | 1 | 10 |
| Q3 | 2 | 1 | 2 | 1 | 1 | 10 |
| Q4 | 2 | 2 | 3 | - | 1 | 10 |
| Q5 | 2 | 2 | 3 | - | 1 | 10 |